# Almost sure non-differentiability of BM.

For a function $f$, define upper & lower right derivatives

$$D^* f(t) = \limsup_{h \downarrow 0} \frac{f(t+h) - f(t)}{h}$$

$$D_* f(t) = \liminf_{h \downarrow 0} \frac{f(t+h) - f(t)}{h}.$$

Theorem (Paley, Wiener, Zygmund 1933)

Brownian motion is nowhere differentiable almost surely.

Further, a.s. for all $t$, either $D^* B(t) = +\infty$ or $D_* B(t) = -\infty$ or both.

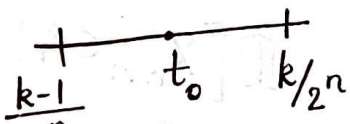
Pf: Suppose $\exists\, t_0 \in [0,1]$ s.t. $-\infty < D_* B(t_0) \leq D^* B(t_0) < \infty$.

$$\limsup_{h \downarrow 0} \frac{|B(t_0+h) - B(t_0)|}{h} < \infty + \{B_t : t \in [0,2]\} \text{ is bounded}$$

$$\Rightarrow \exists\, M < \infty \text{ s.t. } \sup_{h \in [0,1]} \frac{|B(t_0+h) - B(t_0)|}{h} \leq M.$$

[will show that this event has prob 0]

Fix $M > 0$.




$1 \leq j \leq 2^n - k$, $\left|B\left(\frac{k+j}{2^n}\right) - B\left(\frac{k+j-1}{2^n}\right)\right| \leq \left|B\left(\frac{k+j}{2^n}\right) - B(t_0)\right| + \left|B(t_0) - B\left(\frac{k+j-1}{2^n}\right)\right|$

$$\leq \begin{cases} M\left(\frac{k+1}{2^n} - t_0 + t_0 - \frac{k-1}{2^n}\right) \leq \frac{3M}{2^n}, & j = 1 \\ M\left(\frac{k+j}{2^n} - t_0 + \frac{k+j-1}{2^n} - t_0\right) = M\left(\frac{2j+1}{2^n} + \frac{2(k-1)}{2^n} - 2t_0\right) \\ \qquad \leq \frac{M(2j+1)}{2^n}, & j > 1. \end{cases}$$

$$\Omega_{n,k} = \left\{ \left|B\left(\frac{k+j}{2^n}\right) - B\left(\frac{k+j-1}{2^n}\right)\right| \leq M\,\frac{(2j+1)}{2^n} : j = 1(1)3 \right\}$$

$$\mathbb{P}(\Omega_{n,k}) \le \prod_{j=1}^{3} \mathbb{P}\left[\left|B\left(\frac{k+j}{2^n}\right) - B\left(\frac{k+j-1}{2^n}\right)\right| \le \frac{M(2j+1)}{2^n}\right]$$

$$\le \left(\mathbb{P}\left[|N(0,1)| \le \frac{7M}{2^{n/2}}\right]\right)^3 \le \left(\frac{7M}{2^{n/2}}\right)^3.$$

$$\mathbb{P}\left[\bigcup_{k=1}^{2^n-3} \Omega_{n,k}\right] \le \frac{(7M)^3}{2^{n/2}} \quad \text{(summable)}$$

Using Borel-Cantelli,

$$\mathbb{P}\left[\exists\, t_0 \in [0,1] \text{ with } \sup_{h\in[0,1]} \frac{|B(t_0+h) - B(t_0)|}{h} \le M\right]$$

$$\le \mathbb{P}\left[\bigcup_{k=1}^{2^n-3} \Omega_{n,k} \text{ for inf many } n\right] = 0.$$

Finally, let $M \uparrow \infty$ to complete the proof.

Markov Property of BM.

<u>Recall</u>: $\{X_n : n \ge 1\}$ is a MC if $\mathbb{P}[X_{n+1} \in A \mid X_0, \dots, X_n]$
$= \mathbb{P}[X_{n+1} \in A \mid X_n]$.

<u>Q</u>. How to make sense of Markov property when index set is uncountable?

Recasting Markov property.

$$\mathbb{P}[X_{n+1} = x_{n+1}, X_{n-k} = x_{n-k} \mid X_n = x_n]$$

$$= \underbrace{\mathbb{P}[X_{n+1} = x_{n+1} \mid X_n = x_n]}_{\text{future}} \; \underbrace{\mathbb{P}[X_{n-k} = x_{n-k} \mid X_n = x_n]}_{\text{past}}$$

Moral: Given the present, the future & past should be ind.

<u>Filtration</u>. $(\Omega, \mathcal{F}, \mathbb{P})$ probability space.

$\{\mathcal{F}_t : t \geq 0\}$ is a collection of $\sigma$-algebras so that

$$\mathcal{F}_s \subseteq \mathcal{F}_t \text{ for } s \leq t.$$

<u>Example</u>: $\mathcal{F}_t^0 = \sigma(\{B_s : s \leq t\})$.

<u>Lemma</u> $\{B_t : t \geq 0\}$ - sBM. $W_t = B_{t+s} - B_s$

Then $\{W_t : t \geq 0\}$ is sBM ind of $\{B_t : 0 \leq t \leq s\}$.

[Formally, $\forall\, A \subseteq C([0,s])$ mble $\quad \mathbb{P}\left[(W_t) \in B, (B_t)_{t=0}^{s} \in A\right]$

$B \subseteq C([0,\infty))$ mble $\quad = \mathbb{P}\left[(B_t)_{t=0}^{s} \in A\right] \cdot \mathbb{P}\left[(W_t)_{t=0}^{\infty} \in B\right]$]

<u>Pf</u>: Let $\{X_t : 0 \leq t \leq s\}$ & $\{Y_t : t \geq 0\}$ be ind BMs.

$$\mathbb{P}\left[(B_{t_1}, \dots, B_{t_k}) \in A_1 \times \dots \times A_k, (W_{u_1}, \dots, W_{u_\ell}) \in B_1 \times \dots \times B_\ell\right]$$

$$= \mathbb{P}\left[(X_{t_1}, \dots, X_{t_k}) \in A_1 \times \dots \times A_k\right] \mathbb{P}\left[(Y_{u_1}, \dots, Y_{u_\ell}) \in B_1 \times \dots \times B_\ell\right].$$

To complete this proof, note that

$$\sigma(\{f : f(t_1) \in A_1, \dots, f(t_k) \in A_k\}) = \mathcal{B}(C([0,s]))$$

This completes the proof.